



PERCEPT-US: A Multimodal American English Child Speech Corpus Specialized for Articulatory Feedback

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Abstract

We present PERCEPT-US, a multimodal corpus of audio and ultrasound data from 126 American English-speaking children ages 8 to 17. Collected during clinical trials investigating biofeedback in speech therapy for residual speech sound disorder (RSSD), it includes 80 children with no history of speech-language-hearing challenges and 46 children with no challenges other than RSSD. The corpus is balanced by sex (58 females, 68 males) and stratified by age and speech therapy history. Participants completed syllabic, word, and sentence level speech production tasks, totaling 24,699 utterances, with a focus on American English rhotics. The corpus demonstration uses mixed-effects linear regression on segmented acoustic and labeled ultrasound data from 69 children to show that rhotic tongue shape categories significantly predict the acoustics of higher formant frequency values - the first study to demonstrate this acoustic-articulatory relationship in a large pediatric sample.

Index Terms: child speech corpus, ultrasound, acoustic-articulatory relationships, English rhotics

1. Introduction

Speech sound disorder (SSD), which affects a significant portion of school-aged children, has been shown to adversely impact social, emotional, and literacy outcomes [1-3]. Evidence-based treatment methods can alleviate these impacts, but research suggests that roughly 5,000 perceptually accurate productions are necessary for a sound to generalize to spontaneous speech [4], a goal that is challenging to achieve given speech language pathologists' (SLPs) overburdened caseloads [5]. This creates a critical need to develop accessible and reliable resources that allow children to continue therapy outside of direct clinical interactions. Tools that incorporate artificial intelligence (AI) have tremendous potential to extend direct clinical services through guided home practice. In response to this need, our team has created and tested the PERCEPT-US corpus. This multimodal corpus includes simultaneously collected ultrasound images of the tongue and audio recordings from children with no history of speech, language, or hearing challenges and children with residual speech sound disorder (RSSD) affecting the American English rhotic /ɹ/ who have participated in biofeedback speech therapy.

1.1. Motivation

Current AI tools for speech typically classify productions as correct or incorrect and have demonstrated general success[6]. However, they have limited insight into the nature of the adjustment needed to achieve perceptually accurate output. SLPs are trained to provide clients with individualized feedback and phonetic cueing to adjust their speech structures for target sound production. However, there are limits to the guidance SLPs can offer based on auditory perception alone. For instance, perceptually accurate English /ɹ/, which is clinically one of the most challenging sounds to address, can be produced using multiple tongue shapes [7]. Several studies have investigated treatment for RSSD affecting /ɹ/ using ultrasound visual biofeedback, where an ultrasound probe placed beneath the chin allows the SLP and child to view the shape and movements of the tongue during speech therapy. The ultrasound display provides a unique ability to pair the SLP's corrective verbal cues with a visual representation of the target tongue shape to assist the child in adjusting their own tongue placement. Recent studies have shown positive outcomes for speech therapy using ultrasound biofeedback, including an advantage for ultrasound biofeedback over traditional treatment [8-9]. Unfortunately, ultrasound remains out of reach for most SLPs due to the cost of the equipment. AI tools capable of inferring articulatory postures from the acoustic signal thus have the potential to offer two benefits: they could mimic the feedback of an SLP during home practice sessions and could simulate the unique contribution ultrasound provides to SLP-guided therapy.

1.2. Contribution

While the number of child speech corpora has increased in recent years, most feature only audio data paired with transcriptions. Ultrasuite, a multimodal audio and ultrasound corpus collected during speech therapy based in the United Kingdom [10], is the closest comparison to the present American English corpus. We have compiled PERCEPT-US from data collected during four large ultrasound studies over the past decade. The current version of the PERCEPT-US corpus includes approximately 25,000 utterances, making it nearly twice as large the Ultrasuite UK English corpus.

While many articulatory variants of English /ɹ/ are recognized (see detail in section 4), they are commonly grouped

into the broad categories of bunched/tip-down and retroflex/tip-up. Considerable basic science research on the articulatory-acoustic relations of English /ɹ/ has shown a similar acoustic signature across the first three formant frequencies regardless of articulatory position [see review in 11]. Previous work on a small number of adults has demonstrated that English /ɹ/ tongue shapes are acoustically distinct at the level of the 4th and 5th formants, F4 and F5 [12, 13]. As a first step towards acoustic-derived articulatory feedback, the current corpus demonstration investigates these F4 and F5 acoustic-articulatory relationships in children using the ground truth tongue shape labels from the ultrasound data paired with the corresponding audio data.

2. Data Collection

Data were collected in laboratory settings at Haskins Laboratories, Montclair State University, New York University, and Syracuse University under protocols approved by their respective institutional review boards and the Biomedical Research Alliance of New York. Children and parents gave informed assent and consent, with both parties agreeing to future data sharing. Audio recordings were collected at 44.1 or 48 kHz. Ultrasound recordings were collected using a combination of Avermedia, Bandicam or Open Broadcaster Software for screen recording and a GE LogiqE ultrasound with an 8C-RS transducer (Montclair), a Siemens Acuson X300 ultrasound with a C8-5 transducer (NYU) or a C6-2 transducer (Haskins), and a MicrUS EXT-1H, rev.D Beamformer with a MC8-4R20S-3 microconvex transducer (Teled Medical Systems; Syracuse). All children completed the speech production tasks described below while audio and ultrasound data were simultaneously recorded.

2.1. Participant Description

The current version of the corpus includes 126 participants (mean age = 11;10 [years;months], range = 8;8 - 17;8). There are 58 females and 68 males. Two datasets collected earlier focused specifically on the American English /ɹ/ sound, while two more recent datasets include a wider variety of speech sounds with an emphasis on /ɹ/ and /s/ sounds. All speakers were from the Tri-State area (CT-NJ-NY) and spoke a rhotic variety of American English in the home. Participants passed a hearing screening and had no history of speech-language-hearing diagnosis, apart from RSSD for 46 children.

Table 1: *Children in the PERCEPT-US corpus*

	Children with RSSD	Children without RSSD
Number of children	46	80
Females, Males	17, 29	41, 39
Mean age (SD)	10;7 (1;5)	12;7 (2;3)
Age range (yrs; mos)	9;0 - 14;8	8;8 - 17;8

2.1.1. Children with RSSD

Simultaneous audio and ultrasound data were collected from 46 children with RSSD during two different treatment studies for American English /ɹ/: 34 children completed 22 sessions of ultrasound biofeedback treatment, and 12 children received 20 sessions of speech therapy, randomly assigned to conditions with or without biofeedback. Data were collected pre- and post-treatment for the first set of children, while for the second set, data were only collected post-treatment. The greater number of

males in this group (Table 1) is in line with previous research showing greater prevalence of RSSD among boys [14].

2.1.2. Children without RSSD

Among the 80 children without RSSD, 36 participated in a 2-day study of speech and sensory function and 44 children completed a 1-day study.

2.2. Study Protocols

The speech tasks were similar across the four previously mentioned studies. There are six different tasks: one syllable-level, four word-level, and one sentence-level. The syllable and sentence tasks included English /ɹ/ in each utterance. The word level tasks have greater variety: one featured English /ɹ/ in each utterance; one featured English /s/ or /z/; one was specialized to investigate tongue shape complexity; and one was specialized with minimal pairs. Tasks included direct imitation (syllable), picture naming (word-level /ɹ/ and tongue shape complexity), and reading (sentence and word-level /s/-/z/ and minimal pairs).

Table 2: *Speech production tasks with counts of participants and utterance counts in parentheses. Word-TSC represents the tongue shape complexity task and Word-MP represents the minimal pair task.*

Speech Production Task	Children with RSSD	Children without RSSD
Syllable	39 (3240)	77 (3465)
Word - /ɹ/	46 (4000)	78 (3900)
Word - /s, z/	13 (754)	40 (2320)
Word - TSC	34 (1675)	38 (950)
Word - MP	13 (936)	41 (2952)
Sentence	13 (65)	78 (390)
Total Utterances:	10,670	13,977

Because this corpus is compiled from four different studies, there are inevitably some differences in data collection, including for the methods used to ensure consistent alignment of the ultrasound with the participant's speech structures. In the oldest of the four studies (ultrasound biofeedback treatment), small blue stickers were placed on the face and on the ultrasound transducer positioned in a microphone stand under the chin during the video recording. These were used to identify transducer displacement during data processing; misaligned ultrasound frames have been discarded [15]. For the remaining three studies, a plastic 3D printed helmet [16] was used to secure the ultrasound transducer under the child's chin. For the two most recent studies, there is imaging of the children's palate and occlusal plane (n=52; 13 with RSSD, 39 without RSSD). Across all studies, the ultrasound transducer was always positioned to collect midsagittal images of the tongue.

3. Data Preparation

All audio and ultrasound data were collected simultaneously at the task level using various live screen recording software. MP4 files from Avermedia software and MKV files from Open Broadcaster Software were converted in VLC media player to a lossless WAV file and MP4 (H.264 + MP3). Bandicam recordings were collected as WAV and MP4 simultaneously. A team of trained research assistants used the WAV files and Praat [17] TextGrid files to manually transcribe each utterance or verify automated transcription of the orthographic labels. The

Montreal Forced Aligner [18] and the PERCEPT pre-trained acoustic model [19] were used to segment phone boundaries.

Processing of the ultrasound MP4 files occurred in Matlab using the GetContours and SLURP programs [20, 21], which link the timestamps of phone boundaries in Praat TextGrid files to ultrasound image frames. All images within the segments of interest were visually inspected by a team of trained research assistants. In some cases, the most representative image of the target phoneme was selected and manually traced by inserting 16 spline anchor points directly beneath the white line in the image, which represents the boundary with air directly above the tongue surface. In other cases, the SLURP algorithm was used to automatically trace all images with 16 anchor points and the research assistants hand-corrected the traced contours if necessary. The 16 anchor points were used to generate 100 X-Y coordinates evenly spaced along the traced contours.

4. Data Annotation

The data processing previously described creates multiple types of annotations per utterance: audio transcription, phone labeling, and frame-based coordinates from tongue shape tracing. A subset of the corpus also includes perceptually rated accuracy and/or rhotic tongue shape categorization.

4.1. Perceptual Rating Labels

All data was originally collected with perceptual accuracy ratings by the treating clinicians or trained experimenters. A subset of the corpus has been re-rated to establish ground truth perceptual ratings. Some of these ratings were completed by SLPs, while others were obtained from untrained listeners on the Amazon Mechanical Turk crowdsourcing platform. For the crowdsourced ratings, the ground truth label comes from the average across nine listeners, as recommended in [22]. We consider both types of ratings as ground truth labels, although one is clinical and the other is from the general public. Rater type is included in the corpus metadata.

4.2. Tongue Shape Category labels for English /ɪ/

Ground truth labels provided in this corpus have been collected with the goal of articulatory feedback for English /ɪ/ in mind. The 100 X-Y coordinates from the frame-based tongue shape tracing process are quantitative labels that correspond to specific frames of interest within the MP4 files. After tracing was complete, the target frames went through an additional annotation process to provide a qualitative ground truth label of tongue shape category. Currently, 2,698 of the perceptually accurate productions from the syllable-level task have been annotated with these qualitative tongue shape category labels. The raw image was extracted from the MP4 with the two preceding and following frames (five frames per utterance). To obscure participant information, all frame sequences were randomized and presented for coding in a custom PsychoPy [23] program. The raw images were qualitatively coded using the ultrasound-based decision tree proposed by [24], which classifies /ɪ/ tongue shapes into one of five categories using a holistic visual inspection approach [11, 24]. The five tongue shape categories (Mid-Bunched, Front-Bunched, Front-Up, Tip-Up, and Curled-Up) were also collapsed into binary bunched (tongue tip down) and retroflex (tongue tip up) categories to align with previous research with adult tongue shapes [11-13]. For this subset of the corpus, due to the lack of occlusal plane imaging and inability to rotate all images to the same plane, it is unclear whether the tongue tip is up or down

in the Front-Up shape, so the category was excluded. To assess reliability, the dataset was re-coded by the same individual approximately one year after the first round of coding (a practice followed by [24]). Cohen's Kappa [25] was computed to assess intra-rater agreement across the two time points, yielding weighted $k = 0.88$ for classification into five categories and unweighted $k = 0.9$ for binary classification. The raw percentage of intra-rater agreement was 88.66% for the five categories and 95.76% for the binary categories. An additional rater labeled 500/2,698 images (18.53%), yielding weighted $k = 0.9$ for classification into five categories and unweighted $k = 0.88$ for binary classification. The raw percentage of inter-rater agreement was 87.11% for the five categories and 96.91% for the binary categories. The inter and intra rater reliability represents "almost perfect" levels of agreement [26]. There are 2,385 images with multiple ratings that agree.

4.3. Future Corpus Development and Distribution

Development and annotation of the PERCEPT-US corpus is ongoing. Future planned work includes ultrasound image labeling for a subset of the perceptually inaccurate portion of the corpus. Expert clinicians familiar with ultrasound biofeedback will provide ground truth labels related to three tongue-related features: anterior constriction, tongue root retraction, and lowering of the tongue body. Additionally, new audio and ultrasound data will be collected over the next two years through a new clinical trial. The PERCEPT-US corpus will be made publicly available through the PhonBank platform [27] as part of the NIH commitment to data sharing that is Findable, Accessible, Interoperable, and Reusable (FAIR).

5. Corpus Demonstration

To demonstrate a potential application of the corpus, we examined the relationship between acoustics and articulation in the subset of the English /ɪ/ syllable task that has both perceptual ratings and tongue shape category labels (2,385 utterances). Previous research using electromagnetometry [12] and vocal tract Magnetic Resonance Imaging [13] found that the fourth and fifth formant frequencies (F4, F5) differed between bunched and retroflex tongue shapes for English /ɪ/. That work was conducted with 1-2 adult participants. The ground truth tongue shape labeling in our corpus provides the unique opportunity to examine this acoustic-articulatory relationship with a larger sample size ($n=69$) and in child speakers. This demonstration specifically asks whether tongue shape predicts the acoustics of American English /ɪ/ when controlling for phonetic contexts such as onset and vowel context, as well as participant sex and participant group.

5.1. Methods

With the goal of measuring the first through fifth formants (F1-F5), the Voweltine Praat script [28] was adapted to use the burg function to select the optimal LPC settings between 5000 – 7500 Hertz, increasing in 500 Hz steps, when estimating 5.5 formants for each participant. It then selected the steadiest 25 milliseconds (ms) in the segmented /ɪ/ interval by comparing the standard deviation of formant measures across 5-ms windows within the interval. The script measured F1-F5 in the steadiest 25-ms window using the settings that yielded the lowest standard deviation and then calculated the mean formant values in Hz for that window. The Voweltine script was selected for use in this demonstration specifically to address the variability inherent to child speech data while avoiding the

time-intensive process of manually selecting optimal LPC settings for each participant/utterance. It was also adapted to measure higher formants, since F3, F4, and F5 are important for English /ɪ/ and its range of tongue shapes.

Once the formant measures were extracted, the acoustic data was linked to participant demographic data and ground truth tongue shape labels using the tidyverse package [29] in R [30]. This resulted in 1,966 observations of formant measures paired with tongue shape labels. Outliers for formants F1 through F5 were identified and excluded using the rstatix package, which uses the boxplot method [31]. A total of 143 observations (7.3%) were excluded. Three mixed-effects linear regression models using the lme4 [32], lmerTest [33], and emmeans [34] packages were used to evaluate whether binary bunched and retroflex tongue shapes predicted the mean values of F3, F4, or F5 in Hz. The models also included fixed effects of syllable position (onset, non-onset), vowel context (front, non-front), and their interaction with tongue shape, as well as fixed effects of participant sex (female, male), age, and their interaction, as well as group (with/without RSSD). To account for variation across individuals and lexical items, the models also included random intercepts for word and random slopes for participant with intercepts for tongue shape, syllable position, and vowel position.

5.2. Results

The mixed-effects linear regression models revealed statistically significant associations between binary tongue shape labels (bunched and retroflex) and mean F3 and F4 frequencies in Hz when controlling for all other variables. There was no significant association between binary tongue shapes and mean F5 frequencies. These results somewhat substantiate previous research in adults.

For F3 frequencies, there was no significant main effect of bunched and retroflex shapes. There was a significant main effect of vowel context ($\beta = -210.1$, $SE = 50.8$, $p < 0.001$) whereby on average, non-front productions resulted in lower mean F3 Hz values regardless of tongue shape. However, there was a significant interaction between tongue shape and vowel context, such that non-front vowels produced using a retroflex tongue shape resulted in higher mean F3 Hz values relative to those produced with a bunched shape ($\beta = 170.17$, $SE = 63.91$, $p < 0.01$; non-onset was reference level). This significant interaction was unexpected, but it suggests that vowel context is important to consider when analyzing the acoustic signature of different English /ɪ/ tongue shapes. (Figure 1).

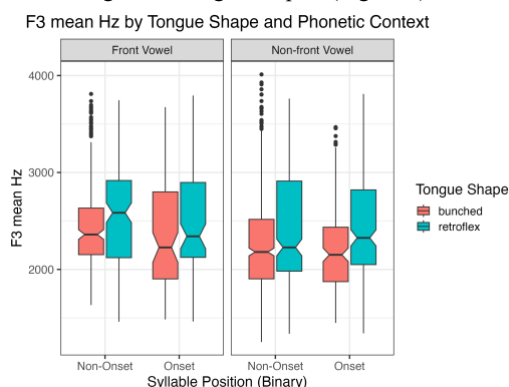


Figure 1: Box plots show mean F3 center frequency, with blue for retroflex shapes and orange for bunched. The x-axis shows syllable position with a facet by vowel position.

Additionally, there were significant associations between binary tongue shapes and mean F4 frequencies. Figure 2 shows that on average, retroflex tongue shapes resulted in significantly lower mean F4 Hz ($\beta = -250.88$, $SE = 100.5$, $p < 0.02$). Vowel context ($\beta = -126.12$, $SE = 47.23$, $p = 0.01$) was also a predictor of mean F4 Hz values.

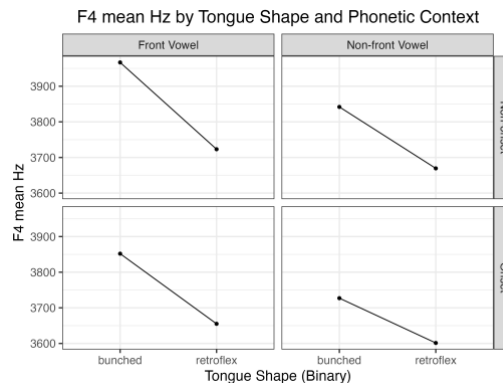


Figure 2: Estimated means plot show the mean F4 center frequency on the y-axis and tongue shape on the x-axis. The plot is faceted vertically by syllable position and horizontally by vowel position.

6. Conclusions and Future Work

This is the first study to examine acoustic-articulatory relationships for American English /ɪ/ in the context of a large sample size and with child speakers. The binary bunched and retroflex tongue shapes predicted F4 frequencies when controlling for all other variables. This was expected based on previous adult research. The significant findings related to F3 frequencies, as well as the lack of significant associations between tongue shape and F5, were unexpected and will be further explored.

Future work aims to train an acoustic classifier to recognize aspects of the tongue shape so that clinicians will be able to "see inside the mouth" (as if they had access to ultrasound imaging) to provide more targeted cueing. Ultrasound remains out of reach for many SLPs and advances in AI have created the potential to develop broadly accessible software that simulates the unique contribution ultrasound provides to speech therapy. In the longer term, we aim to integrate this articulatory feedback technology with our team's accuracy-based classifier (PERCEPT-R)[19] and our team's staRt software for visual-acoustic biofeedback. Software that combines all of these tools is expected to support effective home practice, which could enhance children's learning while reducing the heavy caseload burden faced by practicing clinicians.

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